

Modelling Financial Markets

Chapter 20 (special chapter): Path Signatures and Realised Volatility — a Research Case Study



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Does a Method Built for Freight Rates Improve Volatility Forecasts?

Path Shape and Volatility Forecasts

- A research case study: the method of Gu et al. (2024), published for freight rates at Amazon, transferred to daily realised variance of 50 assets from VOLARE (VOLatility Archive for Realized Estimates) Cipollini et al. (2026)
- Main route: targets and losses, HAR, the path and its worked signature example, kernel weights, then the empirical comparison. The rough-path theory, universality and partial differential equation discussion are further study
- Part II: the paper of Gu et al., a design fixed before the results, results, robustness, one pass of the discovery loop assisted by AI (artificial intelligence)
- Implementation detail sits in the Appendix at the end

What you should be able to do

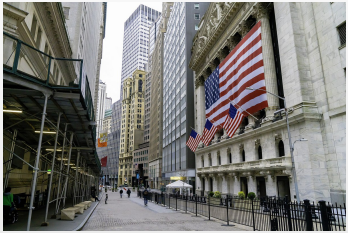
- compute the signature of a piecewise-linear path and read its Lévy area
- build signature features and kernel weights for a volatility forecast
- compare forecasts on many assets with Diebold–Mariano (DM) tests, model confidence sets and Holm and Benjamini–Hochberg (BH) corrections
- pre-register a method transfer and report a null result with its mechanism

Why Forecast Realised Variance?

- Volatility is an input to almost every risk measure in this course
  - Tail-risk measures (Chapter 7), option prices (Chapter 12), portfolio weights (Chapter 4)
- Intraday data make daily variance almost observable: realised variance (RV, Chapter 9)
  - So volatility forecasting becomes a regression problem with a noisy but model-free target
- The benchmark is hard to beat: the HAR (heterogeneous autoregressive) model Corsi (2009) (Chapter 9)
  - Three averages of past RV; linear; estimated by OLS (ordinary least squares) in a second
- Any new method must beat HAR out of sample, on many assets, with a proper test

Two Ingredients to Test: Signature Features and Kernel Weights

- Gu et al. describe a recent window of a series as a **path** and summarise it by its **signature**
  - Signature features enter a sparse linear regression (LASSO, least absolute shrinkage and selection operator, in two steps)
  - Past periods whose path looks like today's get more weight (a signature kernel)
- They report a more than fivefold accuracy gain over an industry forecast of freight rates
  - Question: does the same recipe improve on HAR-type models for realised variance?
- Each ingredient is tested separately, so that a gain can be attributed to the features or to the weights



Wall Street and the New York Stock Exchange, 25 March 2020

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Realised Variance and Quadratic Variation (1/2)

- The price model: the log price moves by a drift, a diffusion term and jumps

$$dp_t = \mu_t\, dt + \sigma_t\, dW_t + dJ_t$$

  - p_t: log price at time t (time in days); μ_t: drift; σ_t: spot volatility; W_t: Brownian motion; J_t: pure-jump process, ΔJ_s = jump size at time s
- Realised variance: the sum of squared intraday returns of day t Barndorff-Nielsen & Shephard (2002)

$$\mathrm{RV}_t = \sum_{i=1}^{M} r_{t,i}^2 \xrightarrow{M\to\infty} \int_{t-1}^{t} \sigma_s^2\, ds + \sum_{t-1<s\le t} (\Delta J_s)^2$$

  - M: number of intraday returns in day t (for a US stock, 78 five-minute returns in the 6.5-hour session); $r_{t,i}$: the i-th intraday log return of day t
  - The limit is the quadratic variation of day t: the integrated variance plus the sum of squared jumps

Realised Variance and Quadratic Variation (2/2)

- Without jumps and noise, on an equally spaced grid, the limit is the integrated variance; the error is mixed Normal

$$\sqrt{M}\,(\mathrm{RV}_t - \mathrm{IV}_t) \Rightarrow \mathrm{MN}(0,\ 2\,\mathrm{IQ}_t)\,, \qquad \mathrm{IV}_t = \int_{t-1}^{t} \sigma_s^2\, ds, \quad \mathrm{IQ}_t = \int_{t-1}^{t} \sigma_s^4\, ds$$

  - IV_t: integrated variance, the true variance of day t; IQ_t: integrated quarticity; $\Rightarrow$: convergence in distribution
  - $\mathrm{MN}(0, V)$: mixed Normal, a Normal with a random variance V; given the volatility path, the error is Normal with variance $2\,\mathrm{IQ}_t/M$

- Reading: RV is a noisy measurement of IV_t, and the noise is largest on volatile days (where IQ_t is large)
  - This is the measurement error that HARQ, the HAR model with realised quarticity, corrects for (next section)
- Microstructure noise makes very fine sampling biased; five-minute returns are the standard compromise (Chapter 9)
- Chapter 9 derives these results; they are used here as given

Semivariances and Quarticity in VOLARE

- Semivariances Barndorff-Nielsen, Kinnebrock & Shephard (2010): RV split by the sign of the intraday returns

$$\mathrm{RV}_t^+ = \sum_{i=1}^{M} r_{t,i}^2 \, 1\{r_{t,i} > 0\}, \qquad \mathrm{RV}_t^- = \sum_{i=1}^{M} r_{t,i}^2 \, 1\{r_{t,i} < 0\}$$

  - $1\{\cdot\}$: indicator, 1 if the condition holds and 0 otherwise; RV_t^+: variance from up-moves, RV_t^-: variance from down-moves
  - They add up: $\mathrm{RV}_t = \mathrm{RV}_t^+ + \mathrm{RV}_t^-$ (holds in VOLARE to machine precision)
- Realised quarticity: an estimate of IQ_t, the scale of the measurement error of RV

$$\mathrm{RQ}_t = \frac{M}{3} \sum_{i=1}^{M} r_{t,i}^4 \;\to\; \mathrm{IQ}_t$$

  - The factor $M/3$: a Normal return with variance s^2 has fourth moment $3s^4$, and each five-minute variance is about IV_t/M
  - The limit needs a continuous, noise-free price and an equally spaced grid; a jump ΔJ adds about $M(\Delta J)^4/3$, so raw RQ is jump-sensitive

Why Five-Minute RV, and Where Do the Data Come From?

- Bipower variation (robust to jumps), realised kernel (robust to noise) Barndorff-Nielsen et al. (2008), range-based estimators
- Target of this chapter: five-minute RV
  - Across 400 assets, little beats five-minute RV as a proxy of daily variance Liu, Patton & Sheppard (2015)
  - Its semivariances and quarticity are on the same grid, so SHAR (the semivariance HAR model) and HARQ use consistent inputs
- Built at the Universities of Messina and Florence from tick data Cipollini et al. (2026)
  - Asset-specific cleaning, trading calendars and timestamp precision; the same estimators for every asset class
- Free for research at volare.unime.it; not redistributed with this course
  - Download the realised-variance files and point the notebooks to your local folder
- Citation requested by the authors
  - Cipollini, Cruciani, Gallo, Insana, Otranto & Spagnolo (2026), arXiv:2602.19732

Targets and Losses (1/2): the Target

- The target at horizon h is the average RV over the next h days

$$\mathrm{RV}_{t+\mathbf{1}:t+h} = \frac{1}{h}\sum_{j=\mathbf{1}}^{h} \mathrm{RV}_{t+j}, \qquad h \in \{1, 5, 22\}$$

  - t: the forecast origin (the last day observed); $h = 1, 5, 22$: one day, one week, one month of trading days
- Log models forecast $y = \ln \mathrm{RV}_{t+\mathbf{1}:t+h}$ and return to levels with a variance correction

$$F_t = \exp(\hat{y}_t + \hat{s}^{\mathbf{2}}/2)$$

  - F_t: forecast of $\mathrm{RV}_{t+\mathbf{1}:t+h}$ made at t; $\hat{y}_t$: fitted log forecast; $\hat{s}^{\mathbf{2}}$: variance of the regression residuals
  - $+\hat{s}^{\mathbf{2}}/2$: since $\mathbb{E}[e^Z] = e^{\mu+s^{\mathbf{2}}/\mathbf{2}}$ for $Z \sim N(\mu, s^{\mathbf{2}})$; exact if the log error is conditionally Normal with constant variance, otherwise an approximation
- For $h > 1$ targets overlap: this induces serial dependence (through lag $h-1$ by construction, possibly longer), so every test uses a HAC (heteroskedasticity and autocorrelation consistent) variance

Matching a Forecast to Its Target

- In the target definition, the subscript names the forecast origin. In loss and test formulas, it labels a matched evaluation row: its realised target and the forecast made before that target became observable.
- Monday close, one-day horizon: use information through Monday to forecast Tuesday RV. Store the Tuesday realisation beside this Monday forecast. Never pair it with a forecast fitted after Tuesday close.
- For a five-day horizon, the same origin forecasts the average over Tuesday through the following Monday (five trading days). Each next origin produces another, overlapping target.
- DM and GW compare two forecasts against the same realised target in each row. The row label is not permission to use that target in estimation.

Targets and Losses (2/2): Robust Losses

- A loss is **robust** if, for a conditionally unbiased proxy, the *expected*-loss ranking with RV_t equals that with the true variance Patton (2011)
  - Conditionally unbiased: $\mathbb{E}_{t-1}[\mathrm{RV}_t] = \sigma_t^2$, with $\mathbb{E}_{t-1}$ the expectation given information up to $t-1$ and σ_t^2 the true variance of day t
- The two robust losses used here (written for $h = 1$; for $h > 1$ replace RV_t by the target)

$$\mathrm{MSE}_t = (\mathrm{RV}_t - F_t)^2, \qquad \mathrm{QLIKE}_t = \frac{\mathrm{RV}_t}{F_t} - \ln \frac{\mathrm{RV}_t}{F_t} - 1 \;\geq\; 0$$

  - F_t: the forecast of RV_t; MSE (mean squared error): the squared error; QLIKE (quasi-likelihood loss) is zero only if $F_t = \mathrm{RV}_t$ and grows faster when the forecast is too low than when it is too high
  - Example with $\mathrm{RV}_t = 1$: $F_t = 0.5$ gives $\mathrm{QLIKE}_t = 2 - \ln 2 - 1 = 0.307$, $F_t = 2$ gives $0.5 - \ln 0.5 - 1 = 0.193$
- QLIKE depends on the ratio RV_t / F_t: the same proportional error has the same loss on a calm day as on a stormy day
- MSE is dominated by a few crisis days; QLIKE is the primary loss, MSE the secondary one; a model is compared through its mean loss over the evaluation days

Crude Oil: Which Prices Enter the Path?

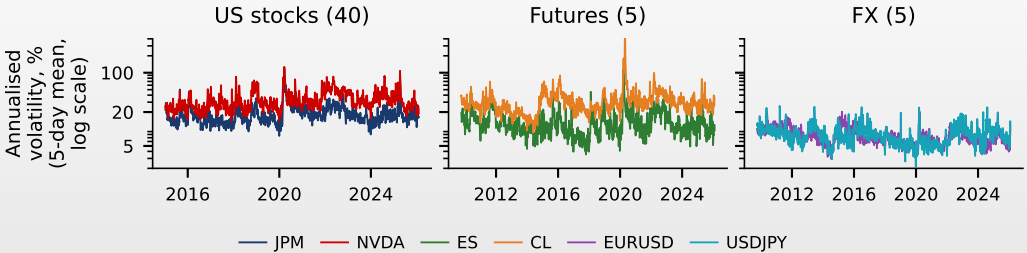
- CL denotes futures on West Texas Intermediate (WTI) crude oil. The negative price in April 2020 concerned one expiring contract. It does not mean that every continuous crude-oil series had negative daily open and close prices.
- The CL series used here is the continuous CL series provided by VOLARE. Its daily open and close prices are positive throughout this sample; on 20 April 2020 they are 23.725 and 21.405.
- A continuous series joins successive contracts as they approach expiry. Its roll rule determines those joins. Compute log returns from these positive VOLARE prices; the negative quote of one expiring contract cannot replace them.

The Panel: 50 Assets in Three Classes

Class	Assets	Sample	Observations	Median volatility
American large-cap stocks	40	2 Jan 2015 – 30 Jan 2026	111 440	17.1%
Futures (C, CL, ES, GC, NG)	5	28 Sep 2009 – 30 Jan 2026	21 059	19.7%
FX (AUD, EUR, GBP, CAD, JPY vs USD)	5	25 Sep 2009 – 30 Jan 2026	21 203	7.3%

- Assets
  - Stocks: 40 members of the Dow Jones 30 and S&P 100, labelled by ticker, from AAPL (Apple) to XOM (Exxon Mobil)
  - Futures: C = corn, CL = crude oil, ES = E-mini S&P 500, GC = gold, NG = natural gas
  - FX (foreign exchange): Australian dollar, euro, British pound, Canadian dollar and Japanese yen against the US dollar
- Median volatility: $\sqrt{252\,\mathrm{median}(\mathrm{RV}_t)}$ per asset, then the median over the class
  - 252 trading days a year turn a daily variance into an annual one; the data contain trading days only
- Variables: five-minute RV, semivariances RV^+, RV^-, realised quarticity (RQ), daily open and close prices

Realised Volatility Across Asset Classes



- Annualised volatility $\sqrt{252\,\mathrm{RV}_t}$ in %, five-day mean, log scale, two assets per class (stocks: JPM = JPMorgan Chase, NVDA = Nvidia; futures: ES = E-mini S&P 500, CL = crude oil; FX: EURUSD = US dollars per euro, USDJPY = yen per US dollar)
- All classes share March 2020; crude oil (CL) peaks in April 2020, when the front contract traded below zero: a forecaster must adapt within days
- On a log scale volatility is stationary and persistent (the long memory HAR approximates); spikes decay within weeks, so the shape of the recent path carries information

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HAR: Three Averages of the Past (1/2)

- The HAR model Corsi (2009): future RV is a linear function of the daily, weekly and monthly averages of past RV

$$\mathrm{RV}_{t+\mathbf{1}:t+h} = \beta_{\mathbf{0}} + \beta_d \,\mathrm{RV}_t + \beta_w \,\mathrm{RV}_t^{(w)} + \beta_m \,\mathrm{RV}_t^{(m)} + \varepsilon_{t+h}$$

$$\mathrm{RV}_t^{(w)} = \frac{1}{5}\sum_{j=\mathbf{0}}^{\mathbf{4}} \mathrm{RV}_{t-j}, \qquad \mathrm{RV}_t^{(m)} = \frac{1}{22}\sum_{j=\mathbf{0}}^{\mathbf{21}} \mathrm{RV}_{t-j}$$

- Notation
  - $\mathrm{RV}_t^{(w)}$, $\mathrm{RV}_t^{(m)}$: weekly (5-day) and monthly (22-day) averages of RV up to day t; ε_{t+h}: forecast error
  - $\beta_{\mathbf{0}}$: intercept; $\beta_d, \beta_w, \beta_m \geq 0$ in practice: how much yesterday, last week and last month weigh; estimated by OLS for each horizon h
  - $\beta_d + \beta_w + \beta_m$ close to 1: high persistence of volatility
- Traders at daily, weekly and monthly frequencies: a cascade that mimics long memory with three parameters

HAR: Three Averages of the Past (2/2)

- HAR is a restricted AR(22): the 22 lag coefficients take only three distinct values

$$\phi_{\mathbf{1}} = \beta_d + \frac{\beta_w}{5} + \frac{\beta_m}{22}, \qquad \phi_{\mathbf{2..5}} = \frac{\beta_w}{5} + \frac{\beta_m}{22}, \qquad \phi_{\mathbf{6..22}} = \frac{\beta_m}{22}$$

  - ϕ_j: the coefficient of $\mathrm{RV}_{t+\mathbf{1}-j}$ in an AR(22) (autoregression with 22 lags); writing out the two averages gives these values

- log-HAR: the same regression in logs

$$\ln \mathrm{RV}_{t+\mathbf{1}:t+h} = \beta_{\mathbf{0}} + \beta_d \ln \mathrm{RV}_t + \beta_w \ln \mathrm{RV}_t^{(w)} + \beta_m \ln \mathrm{RV}_t^{(m)} + \varepsilon_{t+h}$$

  - No negative forecasts, less driven by spikes; back to levels with $F_t = \exp(\hat{y}_t + \hat{s}^{\mathbf{2}}/2)$
  - Logs of averages are not averages of logs, so this log-HAR is not an exact restricted AR(22) in $\ln \mathrm{RV}$ (Seminar 20, A7)

Two Refinements (1/2): HARQ

- RV is a generated regressor: it measures the true variance with an error

$$\mathrm{RV}_t = \mathrm{IV}_t + \eta_t, \qquad \mathrm{Var}(\eta_t) \approx 2\,\mathrm{IQ}_t / M$$

  - η_t: measurement error; errors-in-variables attenuate (pull towards zero) β_d, most on noisy days
- HARQ Bollerslev, Patton & Quaedvlieg (2016): the weight on yesterday's RV falls when RV is measured less precisely

$$\mathrm{RV}_{t+\mathbf{1}:t+h} = \beta_{\mathbf{0}} + \left(\beta_d + \beta_Q(\mathrm{RQ}_t^{\mathbf{1/2}} - \overline{\mathrm{RQ}^{\mathbf{1/2}}})\right)\mathrm{RV}_t + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + \varepsilon_{t+h}$$

  - $\mathrm{RQ}_t^{\mathbf{1/2}}$: square root of realised quarticity, a measure of the error of RV_t; $\overline{\mathrm{RQ}^{\mathbf{1/2}}}$: its in-sample mean, so β_d is the weight on an average day
  - Expected $\beta_Q < 0$: on a noisy day (RQ_t above average) yesterday's RV gets less weight
- BPQ (Bollerslev–Patton–Quaedvlieg) filter: a forecast outside the in-sample range of the target is replaced by the in-sample mean ("insanity filter")

Two Refinements (2/2): SHAR

- SHAR Patton & Sheppard (2015): split the daily term by the sign of the intraday returns

$$\mathrm{RV}_{t+1:t+h} = \beta_0 + \beta^+ \mathrm{RV}_t^+ + \beta^- \mathrm{RV}_t^- + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + \varepsilon_{t+h}$$

  - RV_t^+, RV_t^-: the positive and negative semivariances of day t; β^+, β^-: their weights in the forecast
  - $\beta^+ = \beta^-$ gives back HAR, since $\mathrm{RV}_t^+ + \mathrm{RV}_t^- = \mathrm{RV}_t$
- Patton and Sheppard find $\beta^- > \beta^+$
  - Variance from down-moves predicts more future variance than variance from up-moves: an intraday version of the leverage effect (Chapter 5)
- The signature can capture a related effect from daily data: the Lévy area between returns and log RV (Path Signatures section)

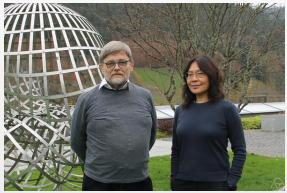
Seven Models in One Table

Model	Target	Weights	Regressors
HAR	RV	equal	$\mathrm{RV}_t, \mathrm{RV}_t^{(w)}, \mathrm{RV}_t^{(m)}$
log-HAR	$\ln \mathrm{RV}$	equal	the same, in logs
HARQ	RV	equal	HAR + $\mathrm{RV}_t \left(\mathrm{RQ}_t^{\mathbf{1/2}} - \overline{\mathrm{RQ}^{\mathbf{1/2}}}\right)$
SHAR	RV	equal	$\mathrm{RV}_t^+, \mathrm{RV}_t^-, \mathrm{RV}_t^{(w)}, \mathrm{RV}_t^{(m)}$
log-HAR-K	$\ln \mathrm{RV}$	signature kernel	log-HAR regressors
Sig-L	$\ln \mathrm{RV}$	equal	log-HAR + 36 non-constant signature terms (39 before dropping constants), two-step LASSO
Sig-LK	$\ln \mathrm{RV}$	signature kernel	as Sig-L: the method of Gu et al.

- Names: L = LASSO, K = kernel weights; the charts call Sig-L "Signature LASSO", Sig-LK "Signature LASSO + kernel weights", log-HAR-K "log-HAR + kernel weights"
- The last four form a 2×2 ablation (each ingredient switched on and off separately): signature features (yes/no) $\times$ kernel weights (yes/no)
- Log-HAR is the natural baseline for the signature models: same target, and its regressors are among the candidates; but the LASSO may also drop log-HAR terms, so Sig-L vs log-HAR compares two procedures, not only two feature sets

Where Signatures Come From

- Chen (1957): iterated integrals of a path form an algebra and determine it up to reparametrisation for the time-augmented paths used here, with starting point fixed
  - Kuo-Tsai Chen, geometry of loop spaces
- Lyons (1998): rough path theory; the signature is the natural feature set for controlled differential equations
  - A response to a path depends on it (to first order) only through its signature
- Machine learning: Levin, Lyons & Ni (2013), Chevyrev & Kormilitzin (2016), Lyons & McLeod (2022); finance: signature-based asset price models Cuchiero, Gazzani & Svaluto-Ferro (2023)



Terry Lyons (left), Oberwolfach, 2024

Photo: Katrin Schmid (2024); CC BY-SA 2.0 DE; Wikimedia Commons

A Window of Data Is a Path

- Take the last L days of $d-1$ series, add time as a channel, and connect the points linearly

$$X : [0,1] \to \mathbb{R}^d, \qquad X_{u_i} = \Big(u_i,\ \ln \mathrm{RV}_{t-L+1+i},\ \textstyle\sum_{j \le t-L+1+i} r_j\Big), \qquad u_i = \frac{i}{L-1},\ i = 0, \dots, L-1$$

  - L: window length in days (here 22); d: number of channels (here $d = 3$: time, log RV, cumulative return); $u_i \in [0,1]$: rescaled time of the i-th point; $r_j = \ln(C_j/O_j)$: open-to-close log return of day j, with O_j, C_j the open and close prices
  - Log returns add up over days, so their sum accumulates the intraday returns in the window, excluding overnight returns; simple changes $C_j/O_j - 1$ compound instead. Example: from 100 to 97 the simple change is -3%, the log return $\ln 0.97 = -3.05\%$; from 100 to 103: $+3\%$ and $+2.96\%$
- **Time augmentation**: the first channel is time itself
  - Without it the signature cannot tell a fast move from a slow one (reparametrisation invariance)
- A regression on the window sees 22 lags; a path summary sees their order, speed and co-movement
  - Example: RV rose while prices fell, or RV rose first and prices fell later

The Signature: Definition (1/2)

- One signature term is an iterated integral of the path along a word of channels

$$S(X)_{a,b}^{i_1 \cdots i_k} = \int_{a<u_1<\cdots<u_k<b} dX_{u_1}^{i_1} \cdots dX_{u_k}^{i_k}$$

- Notation
  - $X : [a, b] \to \mathbb{R}^d$: a path of bounded variation (finite total length), e.g. piecewise linear; X_u^i: its channel i at time u
  - Word $(i_1, \ldots, i_k)$: an ordered list of k channels, letters in $\{1, \ldots, d\}$; k is the **level** of the term
  - $u_1 < \cdots < u_k$: the integral runs over ordered times; dX_u^i: increment of channel i at time u (for a linear segment, slope times du)

The Signature: Definition (2/2)

- The signature is the whole sequence of levels

$$\mathrm{Sig}(X) = (1,\ S^{(\mathbf{1})},\ S^{(\mathbf{2})},\ S^{(\mathbf{3})}, \dots), \qquad S^{(k)} \in (\mathbb{R}^d)^{\otimes k}$$

  - $S^{(k)}$: the d^k terms of level k, one for each word of length k, stored as a k-dimensional array (a tensor; $\otimes$: tensor product)
  - $\mathrm{Sig}^N(X)$: the truncated signature, levels 0 to N; N is the depth
- Level 1: the total increments of the channels

$$S^i = X_b^i - X_a^i$$

- Level 2: increments of channel j weighted by how far channel i has already moved

$$S^{ij} = \int_a^b (X_u^i - X_a^i)\, dX_u^j$$

  - Level 3 adds one more integral; higher levels record finer details of the order of the moves

Linear Segments and Chen's Identity (1/2)

- A linear segment with increment $\Delta \in \mathbb{R}^d$ has the tensor exponential as signature

$$\mathrm{Sig}(\text{segment}) = \exp_{\otimes}(\Delta) = \left(1,\ \Delta,\ \tfrac{1}{2!}\Delta^{\otimes \mathbf{2}},\ \tfrac{1}{3!}\Delta^{\otimes \mathbf{3}}, \dots\right)$$

  - $\Delta = X_b - X_a$: the increment of the segment; $\Delta^{\otimes k}$: the array of all products $\Delta^{i_\mathbf{1}} \cdots \Delta^{i_k}$, so a level-$k$ term is $\Delta^{i_\mathbf{1}} \cdots \Delta^{i_k}/k!$

- **Chen's identity** Chen (1957): the signature of two paths run one after the other is the tensor product of their signatures

$$\mathrm{Sig}(X * Y) = \mathrm{Sig}(X) \otimes \mathrm{Sig}(Y), \qquad (A \otimes B)^{(k)} = \sum_{p=\mathbf{0}}^{k} A^{(k-p)} \otimes B^{(p)}$$

  - $X * Y$: concatenation, Y starts where X ends; $A^{(k)}$, $B^{(k)}$: level k of the signatures A, B, with $A^{(\mathbf{0})} = B^{(\mathbf{0})} = 1$
  - Level k of the whole path collects every way of splitting a word of length k between the first and the second piece

Linear Segments and Chen's Identity (2/2)

- A piecewise-linear path with m segments, increments $\Delta_1, \ldots, \Delta_m$

$$\mathrm{Sig} = \exp_{\otimes}(\Delta_1) \otimes \cdots \otimes \exp_{\otimes}(\Delta_m)$$

  - This is the whole algorithm: m updates, each costing $O(d^N)$ at depth N (a few lines of numpy); $O(\cdot)$: order of the number of operations

- Rolling windows: Chen also removes an initial piece

$$\mathrm{Sig}(X_{[s,t]}) = \mathrm{Sig}(X_{[a,s]})^{-1} \otimes \mathrm{Sig}(X_{[a,t]})$$

  - $X_{[s,t]}$: the path restricted to $[s, t]$, with $a \leq s \leq t$; $(\cdot)^{-1}$: inverse for $\otimes$, the signature of the piece run backwards

Worked Example: a Two-Segment Path

Path

- $X : (0,0) \to (1,0) \to (1,2)$; $\Delta_1 = (1,0)$, $\Delta_2 = (0,2)$
- Level 1: $S^{(1)} = \Delta_1 + \Delta_2 = (1,2)$
- Level 2 by Chen: $S^{ij} = \frac{1}{2}\Delta_1^i\Delta_1^j + \Delta_1^i\Delta_2^j + \frac{1}{2}\Delta_2^i\Delta_2^j$
- Δ_m^i: component i of the increment of segment m; the middle term: channel i moves in segment 1 before channel j moves in segment 2

Result

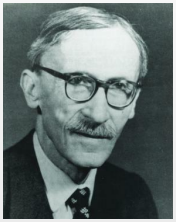
- $S^{11} = \frac{1}{2}$, $S^{12} = 0 + 1 \cdot 2 + 0 = 2$
- $S^{21} = 0$, $S^{22} = \frac{1}{2} \cdot 2^2 = 2$
- Check: $S^{11} = (\Delta X^1)^2/2$ and $S^{22} = (\Delta X^2)^2/2$ (shuffle identity: the product of two level-1 coordinates equals the sum of their two orderings at level 2)
- $S^{12} \neq S^{21}$: the order matters (first right, then up)

The Lévy Area: Order as a Number

- The Lévy area A^{ij}: the antisymmetric part of level 2

$$A^{ij} = \tfrac{1}{2}(S^{ij} - S^{ji})$$

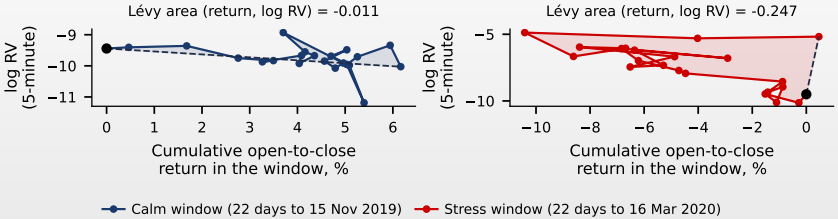
  - Equals the signed area between the path, projected on channels (i, j), and its chord (the straight line from start to end)
  - $A^{ij} > 0$: channel i tends to move before channel j; $A^{ij} = -A^{ji}$
- Symmetric part: $\tfrac{1}{2}(S^{ij} + S^{ji}) = \tfrac{1}{2}\,\Delta X^i\,\Delta X^j$, no new information
  - Example above: $A^{\mathbf{12}} = (2 - 0)/2 = 1$, the area of the triangle under the path
- Channels (return, log RV): a price fall followed by a volatility rise gives $A < 0$, but so does a rise followed by a volatility fall
  - A lead–lag statistic: leverage-type sequences (Chapter 5) are one contributor to a negative area; the sign alone does not identify them



Paul Lévy (1886–1971)

Photo: Konrad Jacobs; CC BY-SA 2.0 DE; Wikimedia Commons

Two Windows of JPM as Paths



- Path (cumulative open-to-close return, log RV) over 22 days; dashed chord from start to end; the shaded region gives the Lévy area
- Calm window (to 15 Nov 2019): log RV moves by -0.58, the path wanders, Lévy area -0.011; stress window (to 16 Mar 2020): log RV rises by +4.32 (RV $\times$ 75) while the intraday return path falls and then recovers, Lévy area -0.247 (returns in decimal units for the area, percent on the axis)
- The net intraday return is only +0.5%: the crash was largely overnight, outside this channel; no signature level recovers what the input omits

MFM_ch20_signatures

Why Signatures Work (1/2): Uniqueness and Shuffles

- **Uniqueness** Hambly & Lyons (2010): the full signature identifies the path
  - $\mathrm{Sig}(X)$ determines X up to tree-like equivalence (adding pieces that are retraced backwards) and translation
  - With a strictly increasing time channel nothing can be retraced, so only translation remains
  - A finite truncation Sig^N is not injective: two different paths can share levels $1..N$
- **Shuffle identity**: a product of signature terms is a sum of higher terms

$$S^i S^j = S^{ij} + S^{ji}, \qquad S^{ij} S^k = S^{ijk} + S^{ikj} + S^{kij}$$

  - The sum runs over all interleavings (shuffles) of the two words that keep the order inside each word
  - So a linear model in $\mathrm{Sig}^N(X)$ already contains polynomials of the increments

Why Signatures Work (2/2): Universality and Decay

- **Universality**: any continuous function of the path is approximately linear in the signature (Stone–Weierstrass, via the shuffle identity)

$$\forall \varepsilon > 0 \;\; \exists\, \ell \text{ linear}: \quad \sup_{X \in K} \left| f(X) - \langle \ell, \mathrm{Sig}(X) \rangle \right| < \varepsilon$$

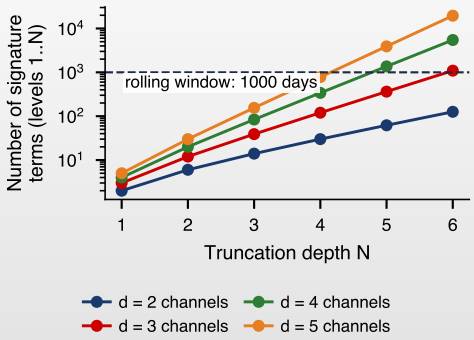
  - K: a compact set of time-augmented paths with a common start (in the 1-variation topology); $f : K \to \mathbb{R}$: a continuous function of the path
  - $\langle \ell, \mathrm{Sig}(X) \rangle$: a weighted sum of finitely many signature terms, i.e. a linear regression on them
  - Sufficient depth permits nonlinear approximation; this does not guarantee accuracy at depth 3 or out of sample. The cost is d^N features, so sparsity is needed

- **Factorial decay**: high levels are small

$$\|S^{(k)}\| \leq \frac{\|X\|_{\mathbf{1\text{-}var}}^k}{k!}$$

  - $\|X\|_{\mathbf{1\text{-}var}}$: total length of the path (sum of the lengths of its segments); $\|S^{(k)}\|$: size (Euclidean norm) of level k
  - Truncating at a small depth N loses little for paths of moderate length, which is why channels are rescaled (Appendix)

How Many Features? The Truncation Dimension



- Number of terms in levels $1..N$ of a d-dimensional path (a geometric sum, since level k has d^k words)
  - $\dim(d, N) = \sum_{k=\mathbf{1}}^{N} d^k = \frac{d^{N+\mathbf{1}} - d}{d - \mathbf{1}}$
- This chapter: $d = 3$ channels (time, log RV, cumulative return), depth $N = 3$: $3 + 9 + 27 = 39$ terms
  - Three are constant (words with only the time letter), so 36 vary
- $d = 5$, $N = 4$ would give 780 terms for 1000 observations: the LASSO would select from noise
- Gu et al. do not disclose N and L; here $N = 3$, $L = 22$ are fixed ex ante, with $N = 2$ as a robustness check

MFM_ch20_signatures

Two Invariances to Keep in Mind

- **Translation invariance**: $\mathrm{Sig}(X + c) = \mathrm{Sig}(X)$ for any constant vector $c \in \mathbb{R}^d$: only increments enter
  - The signature does not see the **level** of volatility, only the shape of its recent path
  - So the level must enter separately: the log-HAR terms play the role of the external factors x_τ of Gu et al.
- **Reparametrisation invariance**: running the path faster or slower, $X_{\psi(u)}$ with ψ increasing, leaves Sig unchanged
  - Removed by the time channel; basepoint augmentation (prepending a segment from 0 to the original starting point) would remove the first invariance Morrill et al. (2020)
  - Translating the path to start at 0 does not: it is itself a translation (Seminar 20, C2 uses the prepended segment)

The Signature Kernel

- Kernel = inner product of feature maps Király & Oberhauser (2019): a similarity score between two paths

$$k_N(X, Y) = \langle \mathrm{Sig}^N(X), \mathrm{Sig}^N(Y) \rangle = 1 + \sum_{k=1}^{N} \sum_{|w|=k} S(X)^w S(Y)^w$$

  - X, Y: two paths with the same channels; w: a word, $|w|$: its length (level); $S(X)^w$: the term of word w; the leading 1 is level 0
- Distance induced by the kernel (Gu et al., Eq. 9): squared Euclidean distance between the truncated signatures

$$d_{\mathrm{Sig}}(X, Y) = k_N(X, X) - 2\, k_N(X, Y) + k_N(Y, Y) = \|\mathrm{Sig}^N(X) - \mathrm{Sig}^N(Y)\|^2$$

  - $d_{\mathrm{Sig}} = 0$: the two windows have the same signature up to depth N; large d_{Sig}: different shapes
- Gu et al. and this chapter use this **truncated** kernel, with N as for the features
  - The untruncated kernel solves a Goursat PDE (partial differential equation) Salvi et al. (2021); the paper cites it but does not use it
- Channels are rescaled before the kernel (Appendix): otherwise the channel with the largest units dominates

The Problem at Amazon and the Features

- Weekly North American truckload rates, national and five regions, 2018–2022, forecast up to 12 weeks ahead Gu et al. (2024)
  - More than 100 candidate factors screened to about 40; in Amazon operations since November 2022; KDD 2024 (Knowledge Discovery and Data Mining, the conference of the ACM, Association for Computing Machinery)
- Features at week τ (Section 4.1, Eq. 14): external factors x_τ and the signature of the time-augmented target path
  - $\xi_\tau = \left[x_\tau,\ \mathrm{Sig}^N(\hat{y}_{\tau-l:\tau})\right], \quad \hat{y}_{\tau-l:\tau} = \{(t_i, y_{t_i})\}$
  - y_{t_i}: the rate in week t_i, $\tau - l \le t_i \le \tau$; l: lookback in weeks; $\hat{y}_{\tau-l:\tau}$: the time-augmented path of the last l weeks
- One direct regression per horizon Δt (in weeks), $y_{\tau+\Delta t}$ on ξ_τ; lookback l, depth N, temperature γ (next slides) are not published (confidentiality)



A container truck on an American interstate, 2025

Photo: DanTD (2025); CC BY 4.0; Wikimedia Commons

The Method: Two-Step LASSO

- Step 1, selection by a weighted LASSO Tibshirani (1996) (Eq. 14)

$$\hat{\theta}_{\mathrm{L}} = \arg\min_{\theta} \sum_{\tau} w_{\tau} \left(y_{\tau+\Delta t} - \xi_{\tau}^{\top} \theta \right)^2 + \lambda \|\theta\|_1$$

  - θ: coefficient vector; $w_{\tau} \geq 0$: weight of week τ (equal, or from the kernel, next slide); $\|\theta\|_1 = \sum_j |\theta_j|$
  - $\lambda \geq 0$: penalty; a larger λ sets more coefficients exactly to zero; $\lambda = 0$ is weighted OLS

- Step 2, refit by weighted OLS on the selected support (Eq. 15)

$$\hat{\theta} = \arg \min_{\mathrm{supp}(\theta) = \mathrm{supp}(\hat{\theta}_{\mathrm{L}})} \sum_{\tau} w_{\tau} \left(y_{\tau+\Delta t} - \xi_{\tau}^{\top} \theta \right)^2$$

  - $\mathrm{supp}(\theta)$: the support, the set of nonzero coefficients; only the regressors kept in step 1 are re-estimated

- Why refit: LASSO shrinks the kept coefficients towards zero; OLS after selection removes that bias Belloni & Chernozhukov (2013)
  - The paper calls it "adaptive" because of the weights w_{τ}, not in the sense of the adaptive LASSO of Zou (2006)
- How λ is chosen is not reported

Choosing the LASSO Penalty in Each Training Window

- The paper does not disclose its penalty selection. Here selection is repeated inside each training window, with no evaluation targets.
- LARS (Least Angle Regression) generates candidate supports along the LASSO path: the solutions as the penalty decreases. Each support is refitted by weighted OLS.
- BIC (Bayesian Information Criterion) balances fit and the number of selected coefficients; the sample-size term uses the Kish effective size $n_{\text{eff}} = 1/\sum_\tau w_\tau^2$, for normalised weights, when weights differ. The smallest BIC selects the support.
- Penalty selection and the BIC formula

The Method: Kernel Weights

- Distance of each past window to the last window with an observed target (Eq. 11)

$$\delta_\tau = d_{\mathrm{Sig},N}\left(z_{\tau-l:\tau},\ z_{t-\Delta t-l:t-\Delta t}\right)$$

  - $z_\tau = (x_\tau, y_{\tau+\Delta t})$: the *pairs* of predictors and later target, not the features ξ_τ (Algorithm 1); $z_{\tau-l:\tau}$: their path over l weeks
  - t: the current week; $t-\Delta t$: the last week whose target y_t is already observed; $d_{\mathrm{Sig},N}$: the signature distance at depth N
- Softmax weights with temperature $\gamma \geq 0$ (Eq. 12): similar windows (small δ_τ) get more weight

$$w_\tau = \frac{\exp(-\gamma\,\delta_\tau)}{\sum_s \exp(-\gamma\,\delta_s)}$$

  - s: runs over all training weeks, so the weights sum to 1; $\gamma = 0$: ordinary (equal-weight) regression; $\gamma \to \infty$: only the most similar past window
- A nearest-neighbour idea inside a regression: learn from the past periods that looked like today
  - Cost of the idea: fewer effective observations, $n_{\mathrm{eff}} = 1/\sum_\tau w_\tau^2$ (Kish); $n_{\mathrm{eff}} = n$ for equal weights $1/n$, $n_{\mathrm{eff}} = 1$ if one week takes all the weight

Three Windows: Weights and Information

- Take distances $\delta = (0.1, 0.4, 0.9)$ and $\gamma = 2$. The unnormalised weights are $(0.819, 0.449, 0.165)$, with sum 1.433.
- Divide by that sum: $w = (0.571, 0.313, 0.115)$. Thus $n_{\text{eff}} = 1/\sum w_i^2 \approx 2.28$, compared with 3 under equal weights.
- The parameter called temperature in the paper acts as inverse temperature: larger γ concentrates weights. The gain is similarity; the cost is less information.
- Try doubling γ: before calculating, predict which weight increases and whether effective sample size rises or falls. Check against the two limits: equal weights and a single selected window.

What the Paper Reports, Read Critically

- National rates, three months ahead, April–November 2021 (Table 1): relative error $|y - \hat{y}|/|y|$ (y: realised rate, $\hat{y}$: its forecast) of about 1–2% vs 13–27% for the industry forecast, "more than fivefold"
  - Horizons 1–12 weeks (Table 2): 1.06–2.60% nationally, 1.05–5.69% by region; estimated annual saving of about 50 million dollars
  - Kernel ablation (Table 3, six weekly values, 31 October–5 December 2021, one region): errors up to 9.85% without the kernel vs up to 5.15% with it
- Limits of the evidence
  - **One benchmark**: an unnamed commercial forecast; no ARIMA (autoregressive integrated moving average) model, no LASSO without signatures, no random walk
  - **Short evaluation**: a few months, a few series, no standard errors, no test of equal accuracy
  - **Undisclosed tuning**: l, N, γ, λ and how they were chosen
- None of this makes the method wrong; it makes the evidence thin, which is why it is tested elsewhere, replication before extension (Chapter 19)

What Transfers to Volatility, and What Does Not

Transfers

- Signature of the time-augmented recent path of the target
- Two-step LASSO with weights, direct regression per horizon
- Softmax weights $\exp(-\gamma\,\delta_\tau)$ with the truncated signature kernel

Must be adapted

- Daily, noisy target instead of weekly rates: logs, HAC inference, robust losses
- External factors x_τ replaced by the log-HAR terms (they carry the level)
- Undisclosed choices (l, N, γ, λ) fixed ex ante in this chapter
- Kernel: the path of this chapter (time, $\ln \mathrm{RV}$, return) ending at τ vs the window ending at t, updated every 5 days, instead of the pairs $(x_\tau, y_{\tau+\Delta t})$ ending at $t - \Delta t$: an adaptation
- Strong benchmarks and formal tests instead of one commercial forecast

Four Hypotheses Fixed Before the Results

- **H1 (features)**: signature terms add information beyond log-HAR
  - Test: Sig-L vs log-HAR, same target, same equal weights
- **H2 (weights)**: kernel weights improve a given regression
  - Tests: Sig-LK vs Sig-L and log-HAR-K vs log-HAR
- **H3 (method)**: the full method of Gu et al. beats the HAR family
  - Tests: Sig-LK vs HAR and vs log-HAR; model confidence set of all seven models
- **H4 (stress)**: any gain is larger in stress periods (COVID-19 crash, 2–30 April 2025, the US tariff shock), where similar past windows should matter most

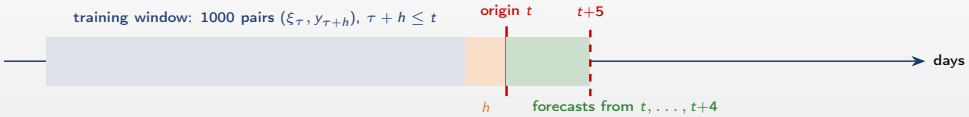
Kernel Temperature: One Choice per Class, on a Validation Block

- Scale-free temperature, set in every training window

$$\gamma = \frac{c}{\mathrm{median}_\tau(\delta_\tau)}$$

  - δ_τ: kernel distances of the training days to the current window; $c > 0$: a dimensionless constant, the only tuning parameter
  - Dividing by the median removes the units of δ_τ; $c = 1$: a window at the median distance gets weight e^{-1} relative to an identical one; larger c: more concentrated weights
- Grid $c \in \{0.5, 1, 2, 4\}$, chosen per horizon and asset class on the first 250 forecast days of each asset, pooled within the class
  - Criterion: mean over the class of QLIKE(c) divided by the asset's average over the grid
  - Evaluation starts after the last validation target of the class, so no tuning target postdates an evaluated origin
- For the 40 stocks the smallest temperature, $c = 0.5$ (the most nearly equal weights), wins at every horizon for Sig-LK; for futures and FX the choice changes with the horizon: a flat, noisy criterion (all values in the Appendix)

Rolling Estimation, Step by Step



- Re-estimate every 5 trading days on the last 1000 complete pairs; between re-estimations the coefficients are fixed but the regressors are updated daily
- Kernel distances are recomputed at each re-estimation, against the window ending at the origin t

Evaluation Protocol (1/2): the DM Test

- Per asset and horizon: mean QLIKE and MSE, compared by the DM (Diebold–Mariano) test of equal predictive accuracy Diebold & Mariano (1995)

$$d_t = L(\mathrm{RV}_t, F_t^A) - L(\mathrm{RV}_t, F_t^B), \qquad \mathrm{DM} = \frac{\bar{d}}{\sqrt{\hat{\Omega}/T}}$$

  - L: the loss (QLIKE or MSE); F_t^A, F_t^B: forecasts of models A and B; d_t: loss differential; $\bar{d}$: its mean over the T evaluation days
  - $\hat{\Omega}$: Newey–West (HAC) long-run variance of d_t, bandwidth $\max(h, \lceil T^{1/3} \rceil)$ lags; $\lceil \cdot \rceil$: rounding up
- Reading: under $H_0 : \mathbb{E}[d_t] = 0$, DM is approximately $N(0,1)$
  - $\mathrm{DM} < -1.645$: model A significantly more accurate at 5% (one-sided); $\mathrm{DM} > 1.645$: significantly less accurate
  - Rolling estimation, so the test is read as the unconditional GW (Giacomini–White) test; the conditional GW test also asks whether d_{t-h} predicts d_t (instruments $(1, d_{t-h})$) Giacomini & White (2006)

From a Loss Difference to Several Tests

- Suppose the mean loss difference (candidate minus benchmark) is -0.02 and its HAC standard error is 0.01. The DM statistic is -2: a one-sided Normal p-value near 0.023 supports the candidate in a single test.
- For three sorted p-values $(0.004, 0.023, 0.08)$, Holm compares sequentially with $(0.05/3, 0.05/2, 0.05)$. The first two pass, the third fails. Stop at the first failure.
- BH compares them with $(0.05/3, 2 \cdot 0.05/3, 0.05)$ and selects the largest passing rank: again two here. Holm controls at 5% the probability of at least one false rejection; BH controls the expected fraction of false rejections under its dependence assumptions.
- Now enlarge the family to 50 tests: these same p-values may no longer pass. The family is a design choice fixed before inspecting the results; Seminar 20 B4 works through the actual asset family.

Evaluation Protocol (2/2): Many Models, Many Assets

- MCS (model confidence set) at 90% Hansen, Lunde & Nason (2011): the set of models that contains the best one with probability at least 90%
  - Built by elimination: while the $T_{\max}$ statistic (the largest standardised gap between a model's mean loss and the average over the models still in the set) rejects equal accuracy, drop the worst model
  - p-values by the stationary bootstrap, 1000 replications, mean block length $\max(h, 10)$ days
- Panel: DM test on the cross-sectional mean loss differential (one series per date)
  - Counting wins over 50 assets: the Holm correction controls the FWER (family-wise error rate, the probability of at least one false rejection), BH Benjamini & Hochberg (1995) the FDR (false discovery rate, the expected share of false rejections among the rejections)
- Subperiods: 20 Feb–30 Apr 2020, 2–30 Apr 2025, all other days; the paper's relative error as a descriptive statistic: definition and results
- Evaluation periods by class and the number of forecasts: ▸ Appendix

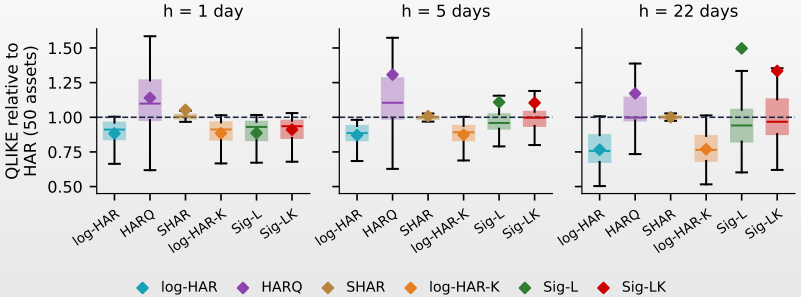
How a Model Confidence Set Shrinks

- Start with three models A, B and C. Suppose the standardised loss gaps are 2.4, -0.5 and -0.7, and the bootstrap p-value for equal accuracy is 0.04. At 10%, reject and remove A, which has the largest standardised gap.
- Recompute the average losses, standard errors and bootstrap test using only B and C. If the new p-value is 0.30, stop: the retained set is B and C. Failure to separate them is not proof that their population losses are equal.
- The nominal 90% coverage is a repeated-sample guarantee under the assumptions of the procedure, not a posterior probability for this particular set. The loss series must be resampled jointly so that dependence across models is preserved.
- Check: if the first bootstrap p-value were 0.14, which models would remain? Explain why selecting only the smallest sample mean is a different rule.

A Disclosed Deviation

- Pre-registered: the BPQ insanity filter on all models
  - A test run of the code on JPM (one asset, before any panel result) showed it replacing correct crash forecasts by the calm in-sample mean
  - On 12 March 2020 the realised next-day RV of JPM was about 40 times the filtered HAR forecast
- Changed before estimating the models on the 50-asset panel: primary results with positivity only (a forecast ≤ 0 becomes the in-sample mean)
  - The BPQ filter is kept as a robustness check and reported in full
- Rule: a deviation is allowed, a hidden deviation is not; state what was seen when it was changed

QLIKE Relative to HAR, Asset by Asset



- Ratio for each asset: mean QLIKE of the model over mean QLIKE of HAR; below 1: more accurate than HAR
  - Boxes: interquartile range over the 50 assets; line: median; diamond: mean; whiskers: 1.5 interquartile ranges

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Averaging Relative Losses Across Assets

- First compute each model mean QLIKE over time for one asset. Divide by the benchmark mean for that same asset. Finally average these ratios over assets, giving each asset equal weight.
- Example: model losses $(0.5, 2)$ and benchmark losses $(1, 0.5)$ give a mean ratio $(0.5 + 4)/2 = 2.25$. The ratio of pooled means is instead $2.5/1.5 = 1.67$.
- Changing the benchmark requires recomputing the per-asset ratios. Dividing the two reported means relative to HAR does not recover the mean ratio relative to log-HAR.
- The same convention applies to the ablation and filtered-results tables. Medians describe a typical asset; means remain sensitive to a few large losses.

Reading the QLIKE Comparison

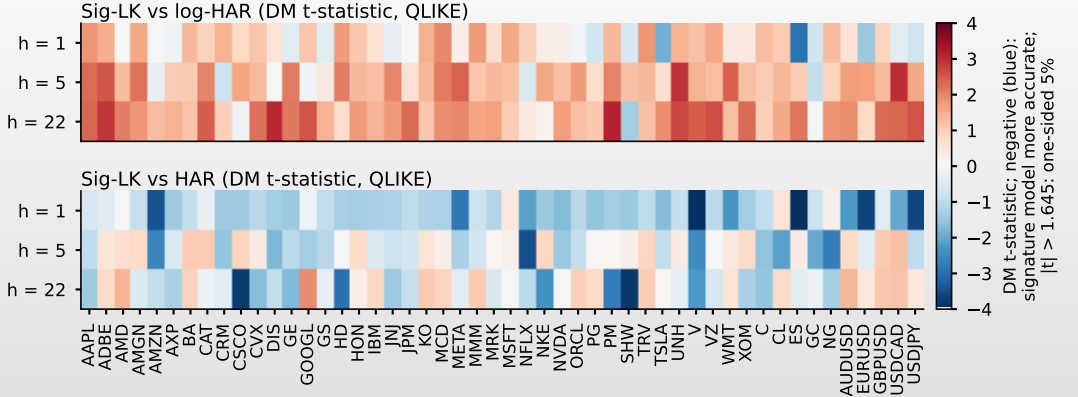
- $h = 1$: log-HAR, log-HAR-K and Sig-L are close (0.883, 0.885, 0.886)
  - Sig-LK is +4.5% relative to log-HAR: no gain from 36 extra features
- $h = 5, 22$: Sig-LK falls behind log-HAR (+28.0%, +92.9%)
  - Its medians relative to HAR (0.997, 0.968) are much better than its means: a few assets and days drive the loss
- HARQ and SHAR lose to HAR in levels here
  - The main design does not apply the BPQ filter (Appendix: the filter rescues HARQ)
- First verdict: the method does not beat the simplest log model
  - A near-tie is not a proof of equal accuracy

Asset by Asset: Wins and Losses, Corrected

Comparison (QLIKE)	$h = 1$	$h = 5$	$h = 22$
Sig-LK vs log-HAR	2/0/0 | 5/0/0	0/0/0 | 15/0/2	0/0/0 | 27/1/16
Sig-L vs log-HAR	3/1/1 | 5/0/0	0/0/0 | 9/0/0	1/0/0 | 21/0/0
log-HAR-K vs log-HAR	3/0/0 | 3/0/0	3/0/0 | 1/0/0	5/0/0 | 6/0/0
Sig-LK vs Sig-L	2/0/0 | 5/0/0	0/0/0 | 5/0/0	2/0/0 | 10/0/0
Sig-LK vs HAR	12/5/6 | 0/0/0	7/1/1 | 0/0/0	8/2/4 | 1/0/0
log-HAR vs HAR	21/5/17 | 0/0/0	23/13/20 | 0/0/0	42/25/42 | 0/0/0

- Cell: significantly better on raw/Holm/BH assets | significantly worse on raw/Holm/BH assets; one-sided DM at 5%, family of 50 assets; Holm controls the FWER under any dependence, BH the FDR under independence or PRDS (positive regression dependence on a subset) Benjamini & Yekutieli (2001)
- H1 (signatures): Sig-L beats log-HAR on 3 of 50 assets at $h = 1$ raw, 1 after Holm; about 2.5 raw wins are expected by chance
- H2 (weights): log-HAR-K vs log-HAR and Sig-LK vs Sig-L: no Holm-significant difference at any horizon
- Against HAR at $h = 1$, Sig-LK wins on 12 assets, but log-HAR alone wins on 21: choose the benchmark that isolates the idea

DM Statistics for Every Asset and Horizon (1/2)



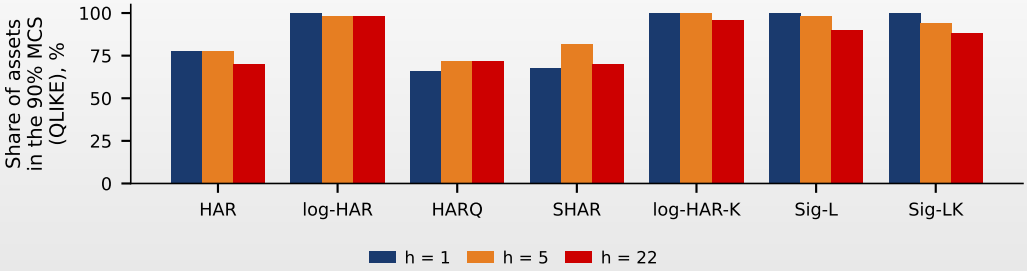
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DM Statistics for Every Asset and Horizon (2/2)

- Diebold–Mariano t-statistics of the QLIKE differential, one row per asset ticker (40 stocks, 5 futures, 5 FX pairs); blue: Sig-LK more accurate, red: less accurate
- Against log-HAR (left): pale at $h = 1$ (blue exceptions ES, TSLA = Tesla, EURUSD; five assets significantly red before correction), red at $h = 5, 22$
- Against HAR (right): blue at $h = 1$ because Sig-LK inherits the log target

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Model Confidence Sets



- Share of the 50 assets in which each model belongs to the 90% MCS under QLIKE: log-HAR 100%, 98%, 98%; Sig-LK 100%, 94%, 88% ($h = 1, 5, 22$)
- The MCS rarely separates the four log models; HAR, HARQ and SHAR are excluded for a fifth to a third of the assets: the level target is the clearest loser
- Under MSE almost every model stays in the set Patton (2011): small differences in average loss need very long samples Pele & Mazurencu-Marinescu-Pele (2026)

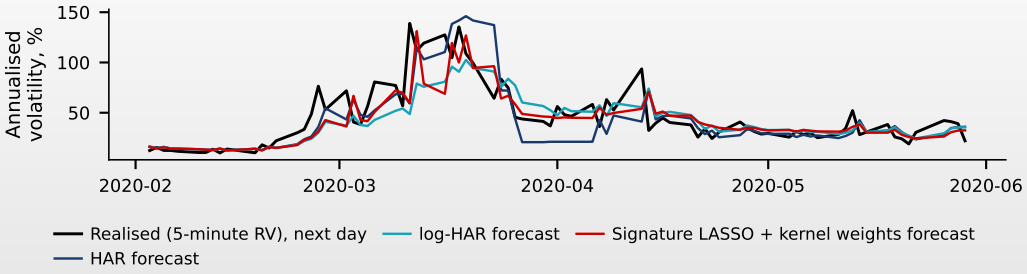
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Panel Tests: Pooling the Evidence

Comparison: DM t (one-sided p)	$h = 1$	$h = 5$	$h = 22$
Sig-LK vs log-HAR	+1.45 (0.927)	+1.68 (0.953)	+1.80 (0.964)
Sig-L vs log-HAR	-1.91 (0.028)	+1.48 (0.931)	+1.51 (0.935)
log-HAR-K vs log-HAR	+1.89 (0.971)	+0.40 (0.654)	+1.92 (0.973)
Sig-LK vs Sig-L	+1.74 (0.959)	+0.98 (0.837)	-0.77 (0.221)
log-HAR vs HAR	-3.09 (<0.001)	-3.95 (<0.001)	-4.50 (<0.001)

- Series: the mean QLIKE differential over all assets trading that day; negative t = first model better
- Pooled, only the log transform is clearly significant; Sig-L vs log-HAR at $h = 1$ is borderline ($p = 0.028$), driven by futures
  - For the 40 stocks alone the sign flips: $t = +1.72$ against Sig-L
- Conditional GW test, Sig-LK vs log-HAR: 0, 0 and 0 Holm rejections out of 50

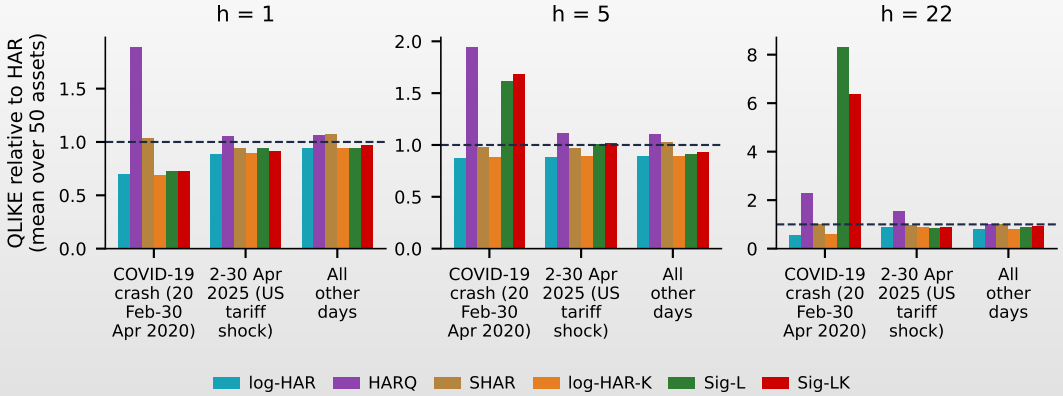
Inside the COVID-19 Crash: JPM, Next Day



- Annualised volatility in % of the forecasts, $\sqrt{252 F_t}$, and of the target, $\sqrt{252\,\mathrm{RV}_{t+\mathbf{1}}}$, February–May 2020; all models lag the first jump
- Sig-LK reacts slightly faster than log-HAR in mid-March, when the path has already turned; after the peak both decay at similar speeds
- HAR in levels overshoots, then turns negative in late March: the positivity rule replaces it by the in-sample mean (the flat segment)

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Stress Periods vs Normal Times (H4) (1/2)



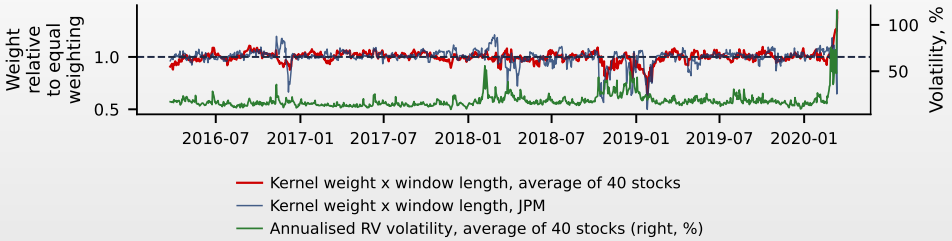
MFM_ch20_case_study

Stress Periods vs Normal Times (H4) (2/2)

- Mean over assets of QLIKE relative to HAR within each subperiod fixed before the evaluation
- $h = 1$, COVID-19 crash: log-HAR 0.70, Sig-LK 0.73, HARQ 1.89; 2–30 April 2025 (US tariff shock): 0.88 and 0.92; other days: 0.94 and 0.97
- $h = 22$, COVID-19 crash: Sig-LK 6.35 and Sig-L 8.30 times the loss of HAR: exactly where similar past windows should help, the signature models fail most; H4 not supported (descriptive, no formal interaction test)

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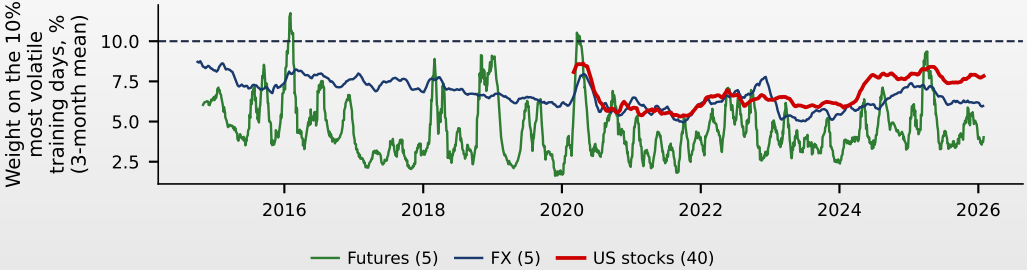
Kernel Weights at the Origin 16 March 2020



- Weights of Sig-LK ($h = 1$) at the origin 16 March 2020 over the training days (24 Mar 2016 – 13 Mar 2020), times the window length: 1 = equal weighting
- Nearly equal: between 0.61 and 1.43 times $1/n$, effective sample size 975 of 1000 (average over the 40 stocks); the highest weights go to the last days before the origin. Recent 22-point windows overlap the current window; similarity may partly reflect overlap, not a distinct economic regime
- The late-2018 sell-off gets *less* weight; weights are almost uncorrelated with log volatility (+0.04): the translation-invariant signature sees shape, not level

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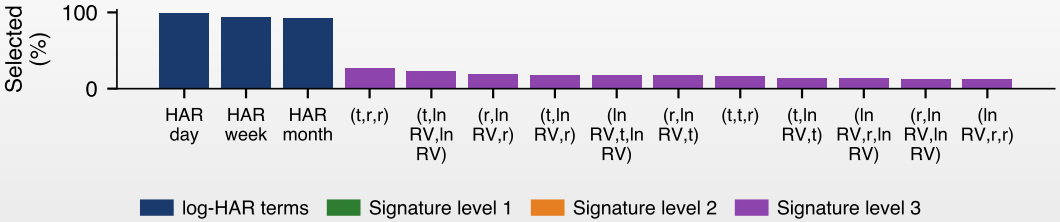
Does the Kernel Ever Favour Stress?



- Weight that Sig-LK ($h = 1$) puts on the 10% most volatile days of its training window, averaged by class, three-month mean
- Stocks: mean 6.7%, between 4.4% and 9.9% (maximum on 17 Mar 2020): always below the 10% of equal weighting; futures and FX show the same pattern
- The mechanism behind the null result for H2: the weights do not implement "learn from past crises"

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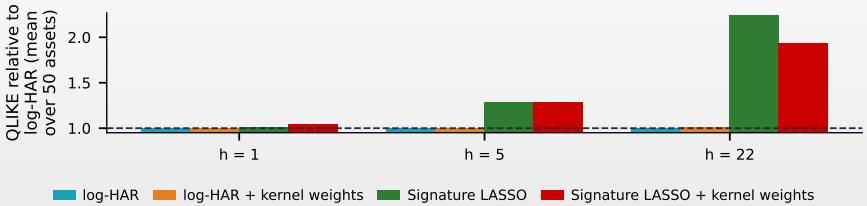
What the LASSO Selects



- Across all assets and horizons: share of re-estimations in which a regressor is in the support of Sig-LK selected by BIC (Bayesian information criterion); labels: HAR day/week/month = the three log-HAR terms, t = time, ln RV = $\ln \mathrm{RV}$, r = cumulative return
- The log-HAR terms are kept in 99%, 94% and 93% of the re-estimations: the LASSO rediscovers HAR; average support 8.7 regressors (Sig-L), 6.7 (Sig-LK)
- The most frequent signature term, (t,r,r) (27%), and all frequent ones are of level 3: cubic in the increments, far outside the training range in a crash, amplified by $\exp(\cdot)$

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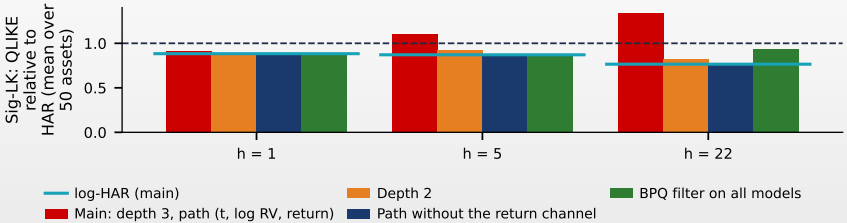
The 2×2 Ablation



- Mean over the 50 assets of QLIKE relative to log-HAR: features (signature or not) crossed with weights (kernel or equal)
- Weights alone (log-HAR-K): 1.002, 1.002, 1.004: no effect
  - Average effective sample size 816 of 1000; features alone (Sig-L): 1.004, 1.285, 2.238
- Both (Sig-LK: 1.045, 1.280, 1.929) damp the one-month failure without removing it
  - They worsen $h = 1$ through crude oil (CL: 2.71); unlike the kernel gain in Table 3 of Gu et al.

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Robustness of Sig-LK



- Sig-LK relative to HAR (mean over 50 assets) in the main design, in two pre-registered variants and with the BPQ filter (the disclosed deviation); the line marks log-HAR
- Depth 2: 0.922 and 0.820 at $h = 5, 22$ (main: 1.104, 1.334); path without returns: 0.878 and 0.770: the long-horizon failures come from level-3 terms with the return channel
- The stable variants at best approach log-HAR: without returns, 0.878 and 0.770 versus 0.871 and 0.765; depth 2 trails it (0.922 and 0.820). Neither change establishes H1; both reduce instability

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By Asset Class

Sig-L relative to log-HAR (QLIKE)	$h = 1$	$h = 5$	$h = 22$
Stocks (40)	1.009	1.332	2.505
Futures (5)	0.965	1.084	1.206
FX (5)	0.996	1.111	1.134

- The only gain: futures at $h = 1$ (-3.5%) and, marginally, FX
  - Five futures and one horizon: a candidate for a new, pre-registered study, not a finding of this one
- Stocks, 80% of the panel, show no gain: the pooled result is mostly a stock result

Limitations and Open Directions

Limitations of this study

- One design out of many ($L = 22$, $N = 3$, three channels, BIC, a four-point grid for c): a null result for this design is not one for signatures in general
- Truncated kernel only; no external factors (implied volatility, macro news), central in Gu et al.
- Realized GARCH Hansen, Huang & Shek (2012) (GARCH: generalised autoregressive conditional heteroskedasticity, Chapter 5) and machine-learning benchmarks (Chapter 13) are outside the comparison; stocks are evaluated over six years only

Open directions

- A level-aware kernel: basepoint augmentation or the level of $\ln \mathrm{RV}$ in the distance (Seminar 20, C2)
- Signatures of the intraday path (5-minute returns); the PDE kernel Salvi et al. (2021); log-signatures and bounded (tanh) channels against explosive level-3 terms Morrill et al. (2020)
- Cross-asset paths, where Lévy areas measure lead–lag across markets; futures at $h = 1$ on a longer panel, with this design fixed in advance

When Does a Method Transfer Across Domains?

- Methods travel faster than evidence: a KDD paper on freight rates becomes a candidate for volatility, credit, energy
  - Which properties of the source problem made the method work?
  - Does the target share them?
- Finance adds a hard filter: 164 teams on the same data disagree more than their standard errors suggest Menkveld et al. (2024)
- Formal version: a transfer succeeds if the gain over the strongest simple benchmark survives a pre-registered set $\mathcal{S}$ of designs, assets and horizons, with the family corrected
  - In this chapter: $\mathcal{S} = 50$ assets $\times$ 3 horizons $\times$ 4 variants; Holm/BH were applied per comparison and horizon (families of 50 assets), variants reported descriptively; no joint correction over all 600 cells; the gain did not survive

The Discovery Loop Applied to This Chapter

- Extraction prompts and falsification prompts provide reusable examples for the steps below.
- **Read**: an assistant lists every equation, hyperparameter and benchmark of the paper and marks what is not disclosed; each item is checked against the PDF
- **Map**: for each component, does it transfer, what must change, which benchmark isolates it
- **Pre-register**: hypotheses, grid, tests and correction fixed in a dated document before the first result; the assistant drafts, the team decides every number
- **Build and test the pieces**: signatures against identities (level 1 = increment, Chen on a split path), not against another library
- **Run the grid, then attack it**
  - which conclusions survive Holm over 50 assets?
  - which single asset would make a misleading headline?
- **Explain the null**: here flat weights (translation invariance) and explosive level-3 terms (exponential link), both checked with data; the researcher's checklist and the full prompts are in the Appendix

Mini-Case: Three Headlines From the Same Runs

- "Signatures beat HAR on 12 of 50 assets" (one-day horizon, raw DM)
  - True, but log-HAR alone beats HAR on 21: the gain belongs to the log transform
- "Signatures improve E-mini S&P 500 volatility forecasts" (ES, $h = 1$, DM $t = -2.91$ against log-HAR)
  - One cell out of 150 asset-horizon cells: after Holm, 0 of 50 at $h = 1$ and none at $h = 5, 22$
- The headline of the pre-registered evaluation: "no gain over log-HAR; worse at longer horizons; kernel weights inert"
  - An AI assistant asked to "find where the method works" will produce the first two; the plan protects the third

Your Turn: Design a Transfer Study

- Pick one open direction of this chapter (level-aware kernel, intraday signatures, log-signatures, futures only)
  - Write H_0/H_1, the benchmark that isolates it, and the result that would make you drop it
- Use an assistant to list what could go wrong in your design, then write one unit test per item
- Fix the pre-registration in a dated document before the first out-of-sample number; run the grid; report the distribution over $\mathcal{S}$
- Log every AI error you catch in `AI_ERRORS.md` (Seminar 20, C1 gives typical ones)

Key Takeaways

- Signatures turn a window into a feature vector with a theory behind it: Chen's identity, uniqueness, universality
- Transferred to realised variance, the method of Gu et al. does not beat log-HAR
  - H1 (signatures add information): not supported, no gain at one day, a loss at a week and a month
  - H2 (kernel weights help): not supported, weights are almost flat and down-weight past stress; H3: only HAR in levels is beaten, through the log target; H4 (gains in stress): not supported
- The translation-invariant kernel sees shape, not level; level-3 terms explode in crashes
- The strongest effect in the study is the oldest one: model the log of RV
- A pre-registered null result, with ablations and a mechanism, is a finding

Readings on Signatures and Realised Measures

Reading

- Signatures: Chevyrev & Kormilitzin (2016); Lyons & McLeod (2022); realised measures: Franke, Härdle & Hafner (2019) and Chapter 9

Advanced courses on quantinar.com

- Statistics of Financial Markets
- Applied Time Series Analysis with Python
- Machine Learning in Financial Risk

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_20; data: VOLARE, free for research at volare.unime.it

Check your understanding

- ⊡ What information about order can a path signature retain that a simple average loses? Use the two-segment example. Revisit.
- ⊡ Why compare with both a level target and a log target? Explain which ablation isolates the contribution of signatures. Revisit.
- ⊡ Can one significant win among many assets establish superiority? Explain the role of correction for multiple comparisons. Revisit.
- ⊡ How can a null result still be informative? Connect the benchmark, the mechanism and the limits of transfer to a new domain. Revisit.

Features and Scaling

Route Decay Kernel

- Regressors at day τ for the signature models: 3 log-HAR terms + 39 signature terms
  - $\xi_\tau = \left[\ln \mathrm{RV}_\tau,\ \ln \mathrm{RV}_\tau^{(w)},\ \ln \mathrm{RV}_\tau^{(m)},\ \mathrm{Sig}^3(X_{\tau-21:\tau})\right]$, path X = (time, $\ln \mathrm{RV}$, cumulative r), 22 points
- Each non-time channel q is multiplied by $\kappa_q = 1/(s_q\sqrt{L-1})$, so its typical total move over the window is about 1
  - s_q: standard deviation of the daily increments of channel q in the training window up to the origin t; $L-1 = 21$ increments per window
  - Time is not rescaled (constant increments, $s = 0$): it runs from 0 to 1; a word $(i_1..i_k)$ is rescaled by $\prod_j \kappa_{i_j}$, so signatures are computed once
- Before the LASSO every column is standardised with weighted moments of the training window; constant columns (words of time only) are dropped
- Raw features at τ use data up to τ; scales and standardisation use data up to the origin t: the no-look-ahead cutoff is t

Estimation: Choosing λ Without Look-Ahead

- Inside each training window: the whole weighted LASSO path by LARS (least-angle regression)
  - For every support on the path: weighted OLS refit (step 2 of Gu et al.)
- Pick the support that minimises a BIC-type criterion (Bayesian information criterion) with the Kish effective sample size, a heuristic of this design
  - $\mathrm{BIC}(S) = n_{\mathrm{eff}} \ln \hat{s}^2_S + (|S|+1) \ln n_{\mathrm{eff}}, \quad n_{\mathrm{eff}} = 1/\sum_\tau w_\tau^2$
  - S: a support (set of kept regressors), $|S|$: its size, $+1$ for the intercept; $\hat{s}^2_S$: weighted residual variance of the refit on S; first term: fit, second: penalty per parameter
- Why not cross-validation: 50 assets $\times$ 3 horizons $\times$ hundreds of re-estimations; BIC uses only the window
  - Kish size corrects for unequal weights, not for serial dependence (overlapping targets, $h > 1$): not a likelihood-based BIC for these data

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Evaluation Windows and Kernel Settings

Class	Evaluation starts	Ends	Days per asset (median)
Stocks	27 Jan 2020	29 Jan 2026	1 499
Futures	25 Sep 2014	29 Jan 2026	2 919
FX	27 Aug 2014	29 Jan 2026	2 952

- 265 929 out-of-sample forecasts per model (50 assets $\times$ 3 horizons)
- Stocks start in 2015, so their evaluation begins in January 2020: the COVID-19 crash is in every stock sample
  - Futures and FX start in 2009 and are evaluated from 2014

- Chosen c for Sig-LK ($h = 1, 5, 22$)
  - stocks 0.5, 0.5, 0.5; futures 2.0, 1.0, 0.5; FX 0.5, 2.0, 0.5
- Chosen c for log-HAR-K ($h = 1, 5, 22$)
  - stocks 0.5, 0.5, 1.0; futures 2.0, 1.0, 4.0; FX 4.0, 2.0, 4.0

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The Paper's Metric on VOLARE

Median of asset mean errors $\lvert \mathrm{RV} - F \rvert / \mathrm{RV}$ (%)	$h = 1$	$h = 5$	$h = 22$
HAR	57.2	48.9	51.7
log-HAR	50.4	41.9	38.7
HARQ	55.0	49.0	50.7
SHAR	57.9	49.2	51.8
log-HAR-K	49.7	41.0	38.7
Sig-L	50.1	42.6	42.6
Sig-LK	50.6	43.6	43.7

- Median over assets of the mean relative error: about 40–60%, against 1–2% for freight rates in Gu et al.
- Daily RV is a noisy, heavy-tailed target: the evaluated models have much larger relative errors than those reported for freight rates, without establishing a lower bound on attainable error
  - Sig-LK forecasts above ten times the in-sample maximum: 8, 29, 36 ($h = 1, 5, 22$), on 10 assets at $h = 5$; log-HAR: none

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The Insanity Filter, Revisited

QLIKE relative to HAR, BPQ filter on all models	$h = 1$	$h = 5$	$h = 22$
HAR	1.000	1.000	1.000
log-HAR	0.763	0.654	0.778
HARQ	0.995	0.912	1.014
SHAR	1.014	1.003	1.000
log-HAR-K	0.772	0.659	0.779
Sig-L	0.871	0.819	0.904
Sig-LK	0.890	0.867	0.930

- The filter rescues HARQ relative to HAR (0.995 at $h = 1$ instead of 1.140), as BPQ intended
- Sig-LK relative to log-HAR, both filtered: 1.238, 1.424, 1.209 (unfiltered 1.045, 1.280, 1.929): the gap widens at $h = 1, 5$ and narrows at $h = 22$
- Filtered forecasts, $h = 1$: HAR 180, HARQ 140, Sig-LK 143, log-HAR 29 (of 89 893): the filter hits the signature models far more often than log-HAR

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What the Researcher Must Check

- **The method is the paper's**: kernel, weights, refit, as in the cited equations
  - Test: every design slide cites an equation, table or section of the paper; undisclosed choices are labelled as choices of the study
- **No look-ahead**: λ, scales, c and features use only data available at the origin
  - Test: change every value after the origin (for example, multiply future RV by 10); the forecast made at the origin must not change
- **The benchmark isolates the idea**
  - Test: the 2×2 ablation is reported, not only the full model against HAR
- **Losses and tests are the right way round**
  - Test: $\mathrm{QLIKE}(\mathrm{RV}, \mathrm{RV}) = 0$ and $\mathrm{QLIKE}(\mathrm{RV}, F) > 0$ otherwise; the DM variance is HAC with bandwidth at least h

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The Discovery Loop, Prompts (1/2)

- **Read**: extract exactly what the paper did
  - "From Gu et al. (2024, KDD), list every equation of the estimator, every hyperparameter and its reported value, and the benchmark of each table. Mark what is not disclosed."
  - Check each item against the PDF: equation numbers, table numbers, the kernel actually used
- **Map**: what transfers, what must be adapted
  - "Our target is daily realised variance. For each component of the method, say whether it transfers, what must change, and which benchmark isolates it."
- **Pre-register**: hypotheses, grid, tests, correction, fixed in a dated document before the first result
  - The assistant drafts; the team decides every number and signs the dated pre-registration

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The Discovery Loop, Prompts (2/2)

- **Build and test the pieces**: signatures against identities, not against another library
  - "Write unit tests for a depth-3 signature: level 1 equals the increment, the symmetric part of level 2 equals half the outer product, Chen's identity on a split path."
- **Run the grid, then attack it**
  - "Here are all 600 runs (50 assets $\times$ 3 horizons $\times$ 4 specifications: main, depth 2, no return channel, BPQ filter). Which conclusions survive Holm over 50 assets? Which single asset or subperiod would make a misleading headline?"
- **Explain the null**: find the mechanism, not an excuse

Back

Acronyms used in this material (1/2)

- **AAPL**: Apple Inc. (Nasdaq ticker)
- **ACM**: Association for Computing Machinery
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive model
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **AUD**: Australian Dollar (ISO 4217 code)
- **BH**: Benjamini–Hochberg (procedure)
- **BIC**: Bayesian Information Criterion
- **BPQ**: Bollerslev–Patton–Quaedvlieg (insanity filter of the HARQ paper)
- **CAD**: Canadian Dollar (ISO 4217 code)
- **CL**: Crude oil (WTI light sweet) futures, NYMEX ticker
- **COVID-19**: COronaVIrus Disease 2019
- **DM**: Diebold–Mariano (test)
- **ES**: E-mini S&P 500 futures (CME ticker)
- **EURUSD**: Euro / US dollar exchange rate (FX pair code)
- **FDR**: False Discovery Rate
- **FWER**: Family-Wise Error Rate
- **FX**: Foreign eXchange
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GC**: Gold futures, COMEX ticker
- **GW**: Giacomini–White (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)

Acronyms used in this material (2/2)

- **HAR**: Heterogeneous AutoRegressive (model)
- **HARQ**: HAR model with Realised Quarticity
- **JPM**: JPMorgan Chase & Co. (NYSE ticker)
- **KDD**: Knowledge Discovery and Data Mining (ACM SIGKDD conference)
- **LARS**: Least-Angle RegreSsion (algorithm)
- **LASSO**: Least Absolute Shrinkage and Selection Operator
- **MCS**: Model Confidence Set
- **MSE**: Mean Squared Error
- **NG**: Natural Gas futures, NYMEX ticker
- **NVDA**: NVIDIA Corporation (Nasdaq ticker)
- **OLS**: Ordinary Least Squares
- **PDE**: Partial Differential Equation
- **PRDS**: Positive Regression Dependence on a Subset (of p-values)
- **QLIKE**: Quasi-LIKElihood (loss function)
- **RQ**: Realised Quarticity
- **RV**: Realised Variance
- **SHAR**: Semivariance HAR model
- **TSLA**: Tesla, Inc. (Nasdaq ticker)
- **US**: United States
- **USDJPY**: US dollar / Japanese yen exchange rate (FX pair code)
- **VOLARE**: VOLatility Archive for Realized Estimates (database)
- **WTI**: West Texas Intermediate (crude oil benchmark)
- **XOM**: Exxon Mobil Corporation (NYSE ticker)

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